

Canine Antibody Production with PawBio™

Advanced Process Development Improves Productivity to Enable Cost-Effective Animal Health Biologics

WuXi Biologics' PawBio service specializes in the research, development, and manufacture of animal health biologics. A critical step in this process is a developability study, which assesses a biologic's commercial potential before investment in clinical trials, manufacturing, and regulatory submissions.

Introduction

A large biopharmaceutical company developing two canine antibodies faced dual challenges of high cost of goods sold (COGS) and low titer. During upstream processing, the low titer was driving up production costs, bioreactor volumes, raw material use, and processing times—inefficiencies that reduced downstream yields and jeopardized commercial success.

Aiming to increase titer and decrease COGS, the company turned to WuXi Biologics for advanced process development. Maintaining the same commercial cell line was a key project requirement.

Challenges

Transferring a project to a contract research, development, and manufacturing organization (CRDMO) carries inherent risks, including quality concerns, higher costs, and extended timelines. As an experienced CRDMO with specialized animal health capabilities, WuXi Biologics quickly onboarded the canine antibody project and turned to focusing process refinements.

The original upstream process delivered a titer of only ~4.5 g/L, requiring yield improvements to support a more efficient downstream process. Downstream processing, however, also presented challenges:

- Small particles during harvest clogged filters, reducing yield and increasing contamination risks.
- Weak Protein A affinity led to poor binding efficiency, greater resin use, and limited scalability. Residual impurities posed quality risks.
- Low viral filtration capacity restricted purification, escalated material costs, slowed processing, and created bottlenecks that compromised viral clearance.



Solution

WuXi Biologics strengthened upstream and downstream processes to enhance productivity, efficiency, and market competitiveness.

Upstream Productivity

The PawBio team synthesized cutting-edge technology with innovative process optimizations to deliver a comprehensive solution. WuXiUI™, WuXi Biologics' technology platform for ultra-intensified fed-batch processing platform, brought a proven track record to upstream development:

- 3–6x productivity increases
- 60–80% lower COGS
- 2–4x higher cell concentrations in only 1.5–3 hours
- Upstream titers of 10–35 g/L for leading CHO cell lines

By surpassing traditional fed-batch processing, WuXiUI significantly reduced COGS and optimized biologic production. In this case study, the technology platform was instrumental in overcoming the upstream titer limitations that affected downstream processing and yield.

Downstream Purification

To ensure high-quality purification during harvest, the PawBio team implemented a two-step purification process. This approach drove measurable gains:

- Eliminated small particles to prevent clogging and contamination
- Strengthened Protein A affinity for better binding efficiency and scalability.
- Increased viral filtration capacity, minimizing bottlenecks and lowering material costs.

Results

The WuXiUI technology platform significantly enhanced upstream productivity, increasing the commercial cell line's titer from ~4.5 g/L to 20.4 g/L - a 450% improvement. Downstream, the two-step purification process boosted overall yields to 65–70%, alleviating bottlenecks and improving efficiency.

The purification solution also provided process performance and product quality attributes comparable to pre-PawBio results, ensuring a consistent, scalable downstream solution. By maintaining the same commercial cell line, the process reinforced continuity and reliability, streamlining integration into future manufacturing efforts.

Conclusion

After a seamless transfer-in, the PawBio team delivered the data to confirm the developability of both canine antibodies and optimize both upstream and downstream processes for a cost-effective animal health solution. Higher upstream titers led to greater downstream yields, and a two-step purification process further improved efficiency.

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About WuXi Biologics

WuXi Biologics is a leading contract research, development, and manufacturing organization (CRDMO) that provides end-to-end capabilities to healthcare organizations worldwide. With operations in China, the United States, Ireland, Germany, and Singapore, we enable our partners to effectively and efficiently bring biologics and vaccines to patients worldwide through our comprehensive and high-quality drug development model.

The world's leading global single-source platform from concept to commercialization

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